Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 11 is the symbol for a nucleus of americium-241.

$$^{241}_{95}\text{Am}$$

Figure 11

Americium-241 is a radioactive isotope of americium.

Americium-241 decays by emitting alpha (α) particles.

(i) Which of these is the symbol for another radioactive isotope of americium?

(1)

☐ **A** $^{241}_{97}\text{Am}$

☐ **B** $^{243}_{96}\text{Am}$

☐ **C** $^{245}_{95}\text{Am}$

☐ **D** $^{247}_{94}\text{Am}$

(ii) Which of these is the approximate maximum distance that alpha particles can travel in air at normal atmospheric pressure?

(1)

☐ **A** 5 mm
☐ **B** 5 cm
☐ **C** 5 m
☐ **D** 5 km

(iii) Complete the equation in Figure 12 for americium-241 decaying into neptunium (Np).

(3)

$$^{241}_{95}\text{Am} \rightarrow \underline{\quad\quad}\ \alpha + \underline{\quad\quad}\ \text{Np}$$

Figure 12

(Total for question = 5 marks)

Q2.

Figure 1 is a diagram of three atoms.

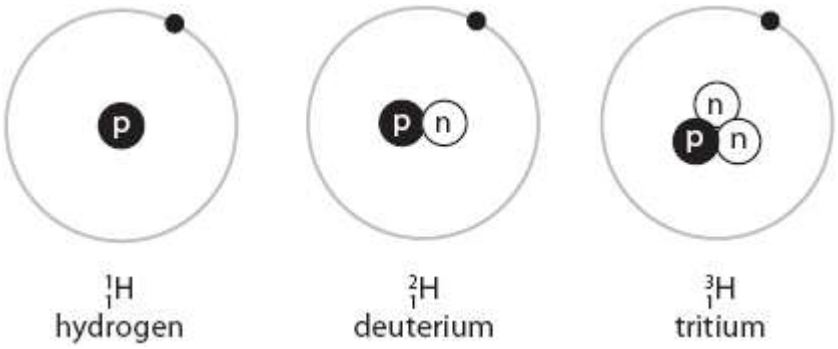

Figure 1

Give reasons why these atoms are isotopes.

(2)

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(Total for question = 2 marks)

Q3.

This question is about radioactivity.

Figure 3 shows a Geiger-Muller (G-M) tube attached to a counter.
The G-M tube is used to measure the activity of a source of beta (β) radiation.
There is an aluminium sheet between the beta source and the G-M tube.
The counter is switched on and after 1 minute shows a count of 268.

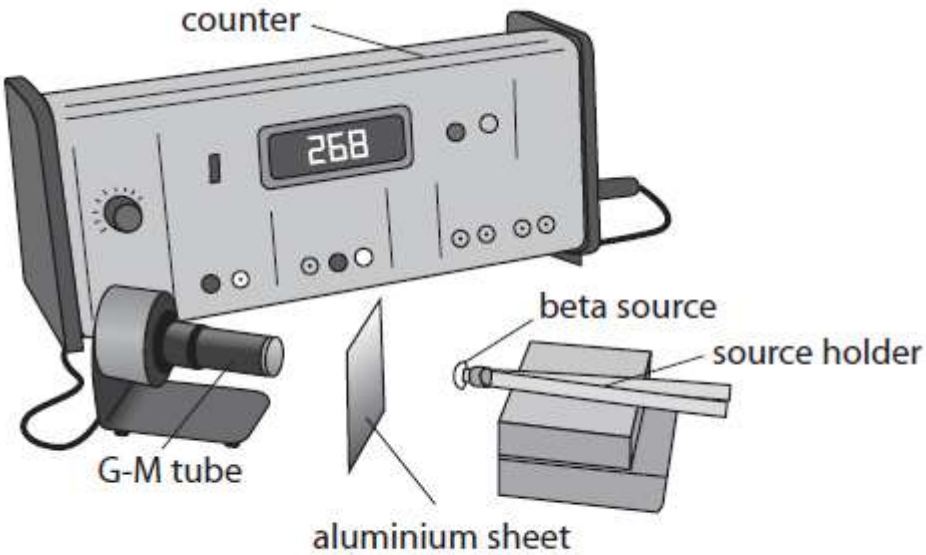

Figure 3

(i) The aluminium sheet is taken away.

The counter is reset to zero and then switched on again.
A new count is taken for 1 minute.

Explain why the new count is greater than 268.

(2)

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(ii) The beta source is then also taken away.
The counter is reset to zero and switched on again.
A new count is taken for 1 minute.

Give a reason why there would now be a reading on the counter.

(1)

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(iii) State the SI unit for the activity of a radioactive source.

(1)

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(Total for question = 4 marks)

Q4.

Fusion and fission are nuclear reactions in which large amounts of energy are released.

(i) In a fusion reaction, two hydrogen nuclei are forced together to form a helium nucleus.

Explain why a very high temperature is needed for this reaction to happen.

(3)

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(ii) In a fusion reaction, the combined mass of the two small nuclei is greater than the mass of the resulting nucleus.

This decrease in mass, m, appears as energy, E, according to the equation.

$$E = mc^2$$

c is the speed of light = 3.0×10^8 m/s.
The energy released in one fusion reaction is 4.5×10^{-12} J.
Calculate the decrease in mass.

(3)

decrease in mass = ... kg

(Total for question = 6 marks)

Q5.

Nuclear fusion is another type of nuclear reaction.

Explain why there are no power stations using nuclear fusion.

(3)

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(Total for question = 3 marks)

Q6.

Ultraviolet (UV) waves from the Sun travel towards the Earth.

Ultraviolet waves can be grouped by wavelength.

The three groups of wavelengths are UVA, UVB and UVC.

Figure 7 shows, for each group,

- the wavelength range
- the effect of the Earth's atmosphere on each type of UV wave.

	UVA	UVB	UVC
wavelength range	400 nm to 315 nm	314 nm to 280 nm	279 nm to 100 nm
% energy absorbed by the Earth's atmosphere	5%	95%	100%

Figure 7

UV radiation of wavelength 365 nm is used to detect forged banknotes.

In a genuine banknote there are marks that **cannot** be seen using visible light.

These marks **can** be seen using UV radiation.

Explain why the marks can be seen when the UV radiation shines on the banknote.

Your answer should refer to the energy of electrons in atoms.

You may draw a diagram to help with your answer.

(4)

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(Total for question = 4 marks)

Q7.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

This question is about using radioactivity and nuclear energy.

(i) Gamma ray irradiation of food kills harmful bacteria by using

(1)

- ☐ **A** ionising particles
- ☐ **B** ionising radiation
- ☐ **C** non-ionising particles
- ☐ **D** non-ionising radiation

(ii) Which part of a nuclear reactor is used to absorb neutrons?

(1)

- ☐ **A** control rods
- ☐ **B** coolant
- ☐ **C** moderator
- ☐ **D** PET scanner

(Total for question = 2 marks)

Q8.

Other unstable isotopes emit alpha particles.

Which of these describes an alpha particle?

(1)

- ☐ **A** a hydrogen nucleus
- ☐ **B** a hydrogen atom
- ☐ **C** a helium nucleus
- ☐ **D** a helium atom

(Total for question = 1 mark)

Q9.

What is the approximate size of a hydrogen atom?

(1)

- ☐ **A** 10^{-3} m
- ☐ **B** 10^{-10} m
- ☐ **C** 10^{-19} m
- ☐ **D** 10^{-31} m

(Total for question = 1 mark)

Q10.

An electron has a charge of −1.

The charge on an alpha particle is

(1)

- ☐ **A** −2
- ☐ **B** 0
- ☐ **C** +1
- ☐ **D** +2

(Total for question = 1 mark)

Q11.

Nuclear fusion is a process that releases energy.

Which of these statements applies to a nuclear fusion reaction?

(1)

- ☐ **A** it emits daughter nuclei
- ☐ **B** it is a controlled chain reaction
- ☐ **C** it produces radioactive waste
- ☐ **D** it requires high temperature and pressure

(Total for question = 1 mark)

Mark Scheme

Q1.

	Answer	Mark
(i)	C $^{245}_{95}Am$ **A, B** and **D** are incorrect as these are not isotopes of americium.	(1) AO1

	Answer	Mark
(ii)	[x] **B** 5 cm **A, C** and **D** are incorrect as these are not the correct range of an alpha particle in air.	(1) AO1

	Answer	Additional guidance	Mark
(iii)	$^{4}_{2}\alpha$ (1) ^{237}Np (1) ^{93}Np (1)	both correct for the mark ecf from mp1 ecf from mp1	(3) AO2

Q2.

Question number	Answer	Mark
	one from • same atomic number (1) • same number of protons (1) • same element (1) and one from • different numbers of neutrons (1) • different mass numbers (1)	(2)

Q3.

Question number	Answer	Additional guidance	Mark
(i) CS2	An explanation to include; there is no aluminium to absorb β particles (1) (therefore) more β particles reach the G-M tube (1)	aluminium absorbs/stops/blocks beta particles accept reverse arguments accept radiation for beta particles	(2) AO2

Question number	Answer	Additional guidance	Mark
(ii) CS2	(idea of) background radiation	a named source of background radiation	(1) AO3

Question number	Answer	Additional guidance	Mark
(iii) CS2	becquerel	accept Bq accept close spelling	(1) AO1

Q4.

Question Number	Answer	Additional guidance	Mark
(i)	an explanation linking: (high temperature means) high energy (1) (needed) to overcome (force of) repulsion (1) between nuclei / because they both have the same charge (1)	accept "them" / hydrogen for nuclei	(3) AO 2 1

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) $4.5 \times 10^{-12} = m\ (3.0 \times 10^{8})^2$ rearrangement (1) $(m =)\ \frac{4.5 \times (10^{-12})}{9.0 \times (10^{16})}$ evaluation (1) 5.0×10^{-29} (kg)	accept substitution and rearrangement in either order ignore POT errors until evaluation award full marks for the correct answer with no working	(3) AO 2 1

Q5.

Question number	Answer	Additional guidance	Mark
	An explanation that makes reference to identification - knowledge (2 marks) and reasoning/justification - knowledge (1 mark): • fusion requires high temperatures and pressures (1) • (in order to overcome) electrostatic repulsion of nuclei (1) • (we are currently) unable to maintain these conditions (in a commercial reactor) (1)		**(3)**

Q6.

Question	**Answer**	**Additional guidance**	**Mark**
	an explanation linking any **four** of the following:- UV/energy absorbed by electrons (1) electrons change orbit/energy level/shell (1) electrons in an 'excited' state (1) electrons emit energy/ move to lower energy level/fall down/de-excite (1) (energy) emitted as (visible) light / at a different frequency/wavelength (1) (process is called) fluorescence / (light emitted is) fluorescent (1)	a labelled diagram can score up to 4 marks	**4** **AO1.1**

Q7.

Question number	Answer	Mark
(i)	B ionising radiation (1) A is incorrect because gamma rays are not considered as particles in this context C and D are incorrect: the gamma rays are ionising, not non-ionising	(1) AO1.1

Question number	Answer	Mark
(ii)	A control rods (1) B,C and D do not absorb neutrons	(1) AO1.1

Q8.

Question number	Answer	Mark
	C a helium nucleus	(1)

Q9.

Question number	Answer	Mark
	B 10^{-10} m	(1)

Q10.

Question	Answer	Mark
	D +2 A , B and C do not apply to an alpha particle	**1** **AO1.1**

Q11.

Question	Answer	Mark
	D it requires high temperature and pressure **A** no emission of daughter nuclei **B** not a chain reaction **C** does not produce radioactive waste	**1** **AO1.1**